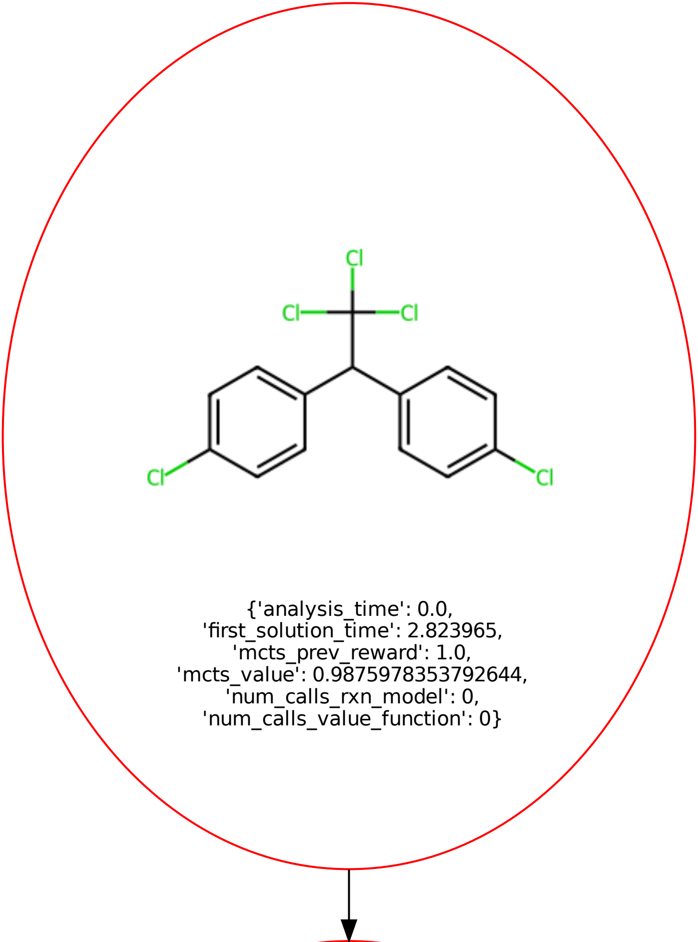
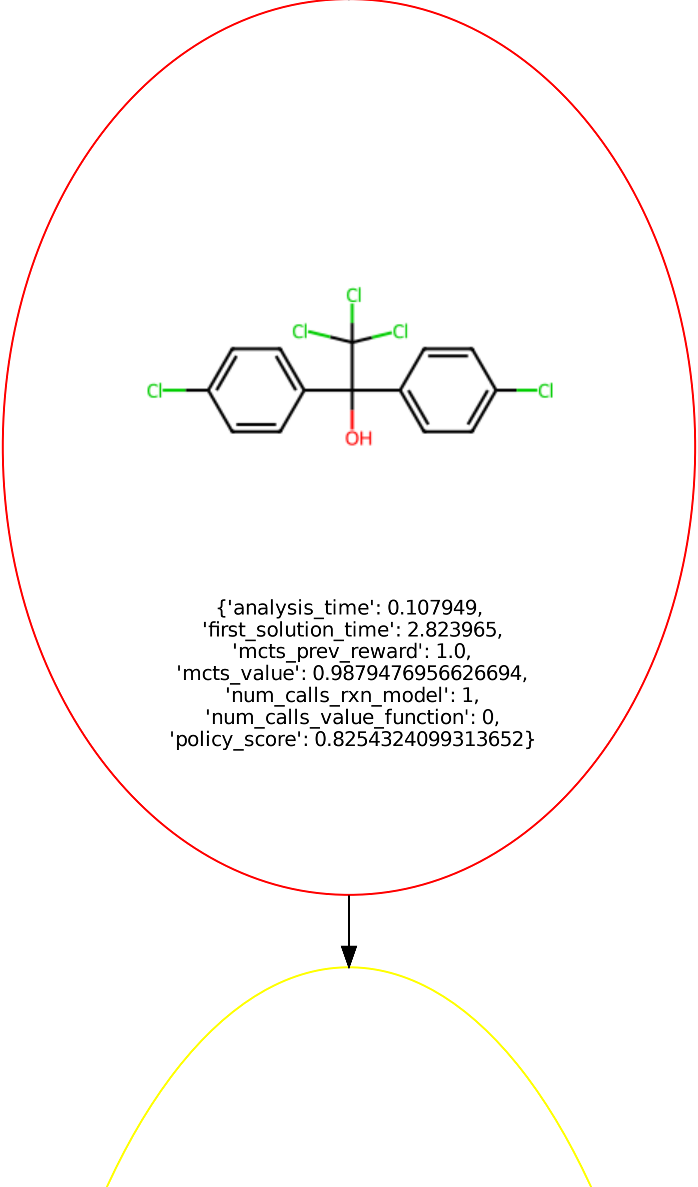
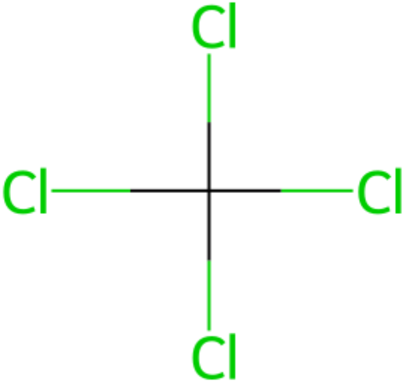
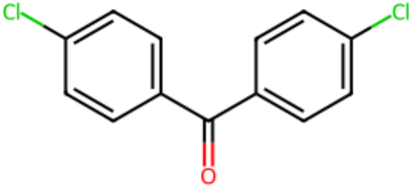


```
{'analysis_time': 0.0,
'first_solution_time': 2.823965,
'mcts_prev_reward': 1.0,
'mcts_value': 0.9875978353792644,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0}
```

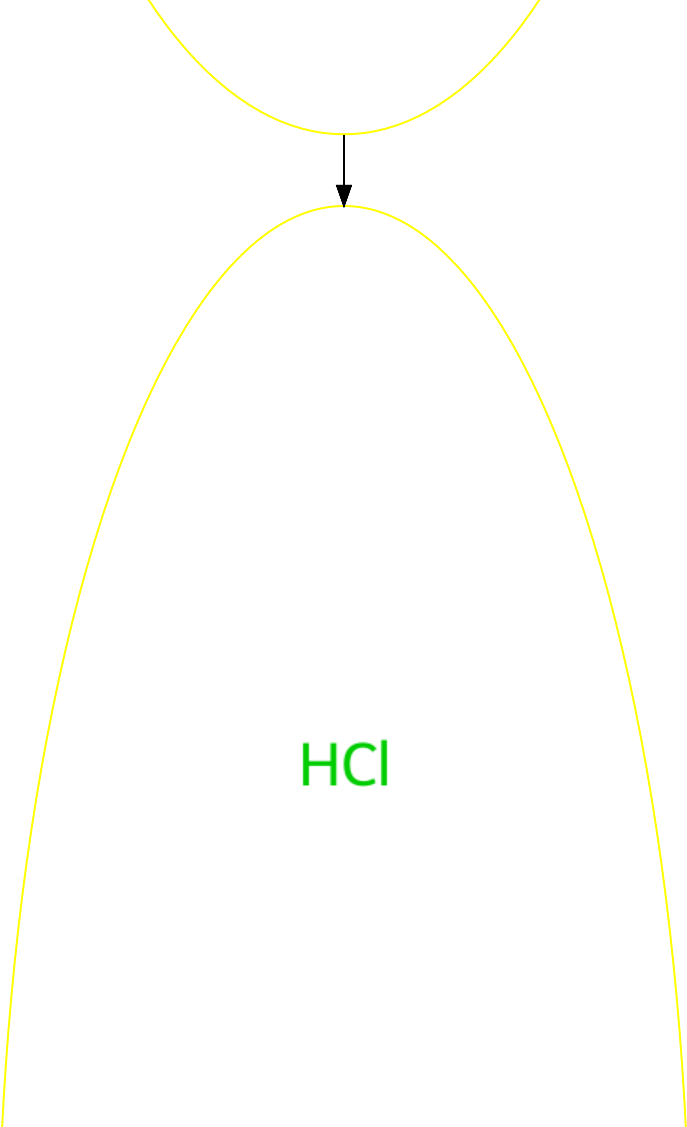


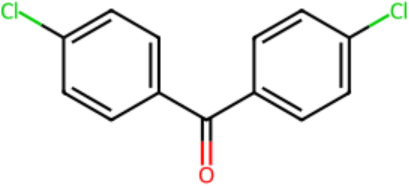
```
{'analysis_time': 0.107949,
'first_solution_time': 2.823965,
'mcts_prev_reward': 1.0,
'mcts_value': 0.9879476956626694,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'policy_score': 0.8254324099313652}
```

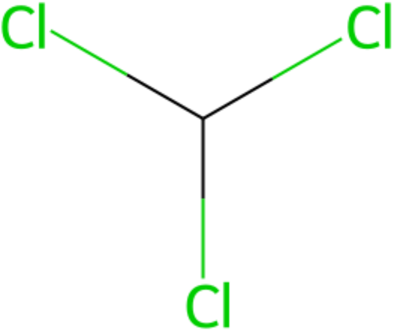




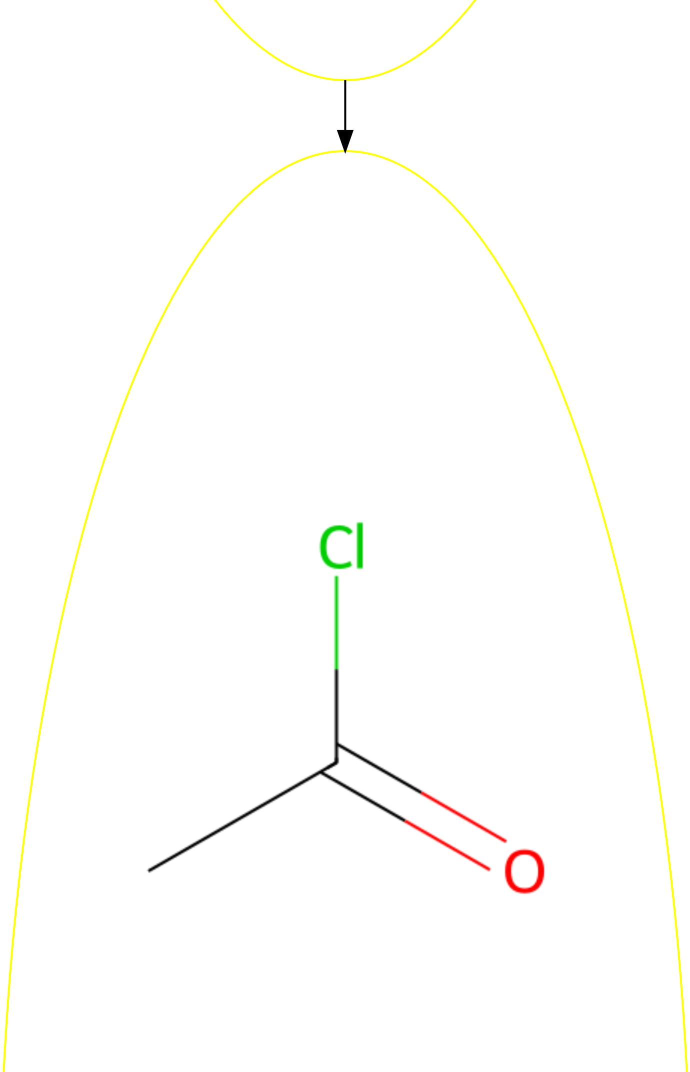
{'analysis_time': 0.267791,
'first_solution_time': 2.823965,
'mcts_prev_reward': 1.0,
'mcts_value': 0.9883645570675392,
'num_calls_rxn_model': 2,
'num_calls_value_function': 1,
'policy_score': 0.7957781537272515}

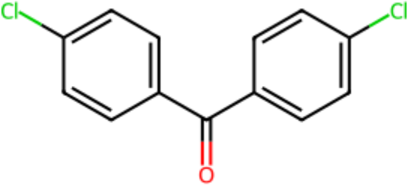


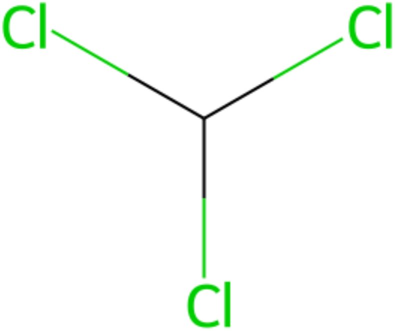




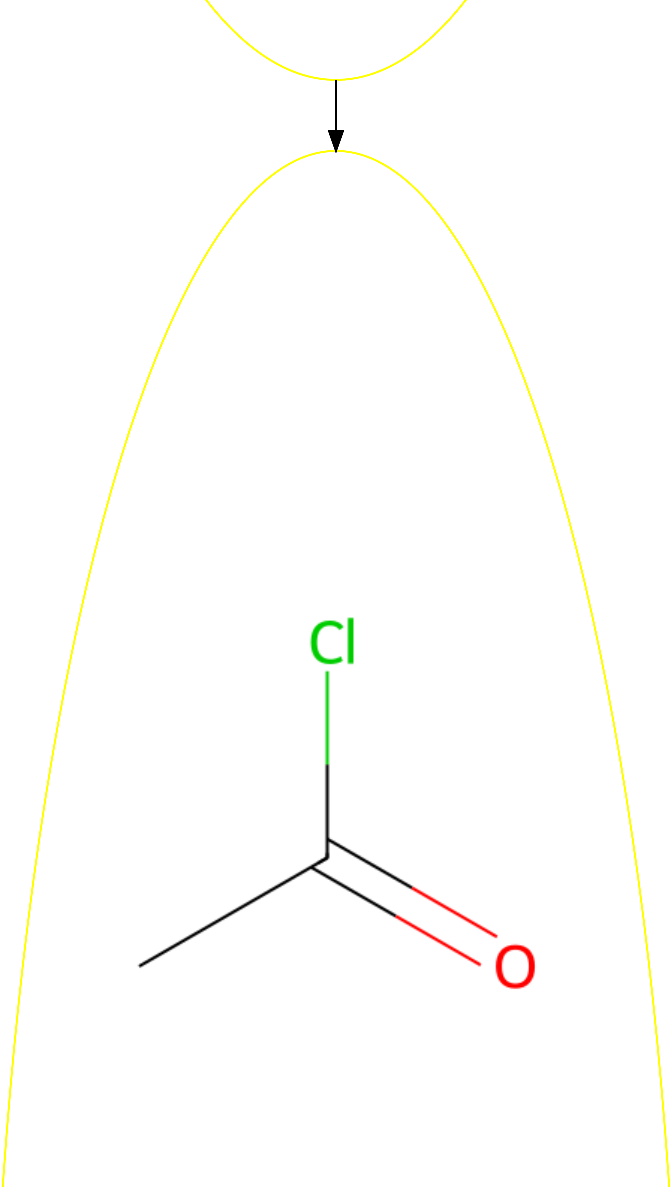
```
{'analysis_time': 0.3764,
'first_solution_time': 2.823965,
'mcts_prev_reward': 1.0,
'mcts_value': 0.9886979824774274,
'num_calls_rxn_model': 3,
'num_calls_value_function': 2,
'policy_score': 0.8343817868209729}
```



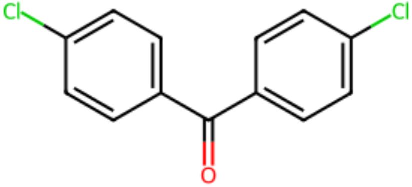


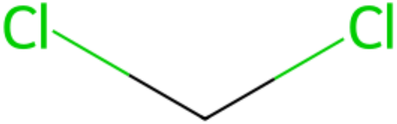


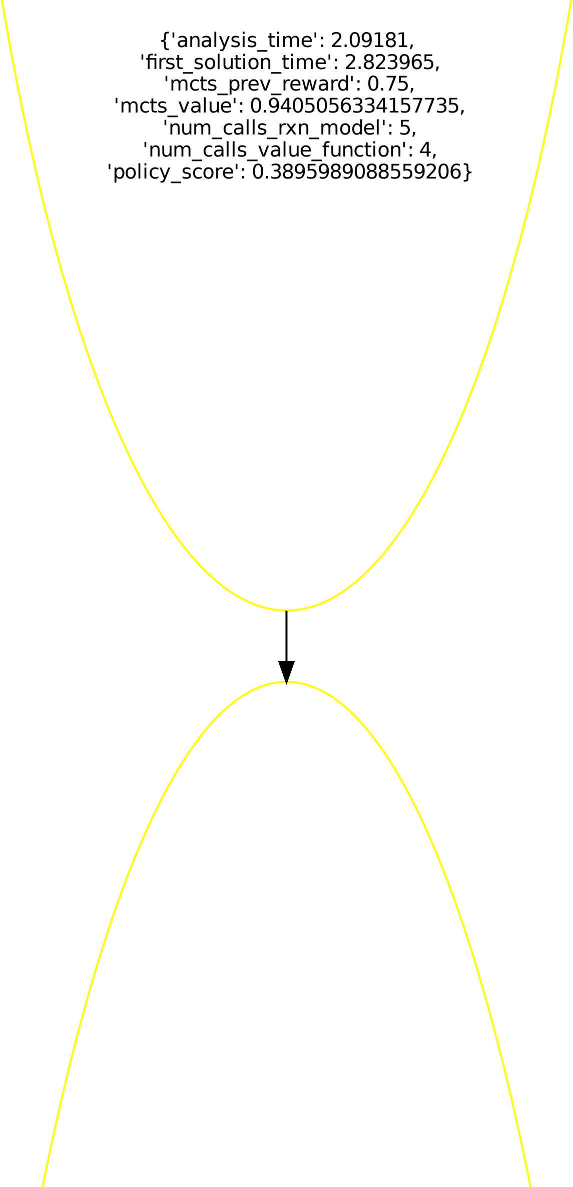
```
{'analysis_time': 2.090649,
'first_solution_time': 2.823965,
'mcts_prev_reward': 1.0,
'mcts_value': 0.9904085721198128,
'num_calls_rxn_model': 5,
'num_calls_value_function': 3,
'policy_score': 0.15847409713933694}
```



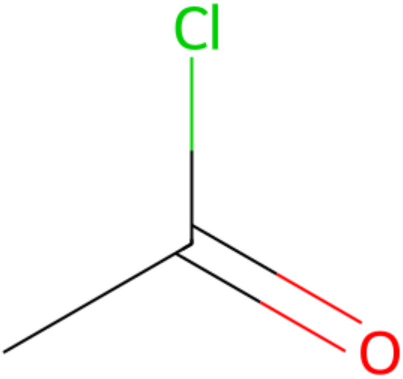
HCl



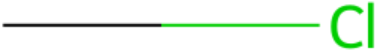


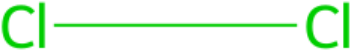


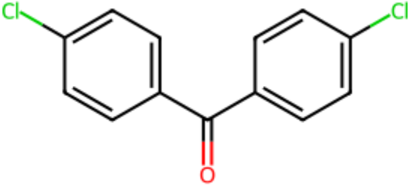
```
{'analysis_time': 2.09181,
'first_solution_time': 2.823965,
'mcts_prev_reward': 0.75,
'mcts_value': 0.9405056334157735,
'num_calls_rxn_model': 5,
'num_calls_value_function': 4,
'policy_score': 0.3895989088559206}
```

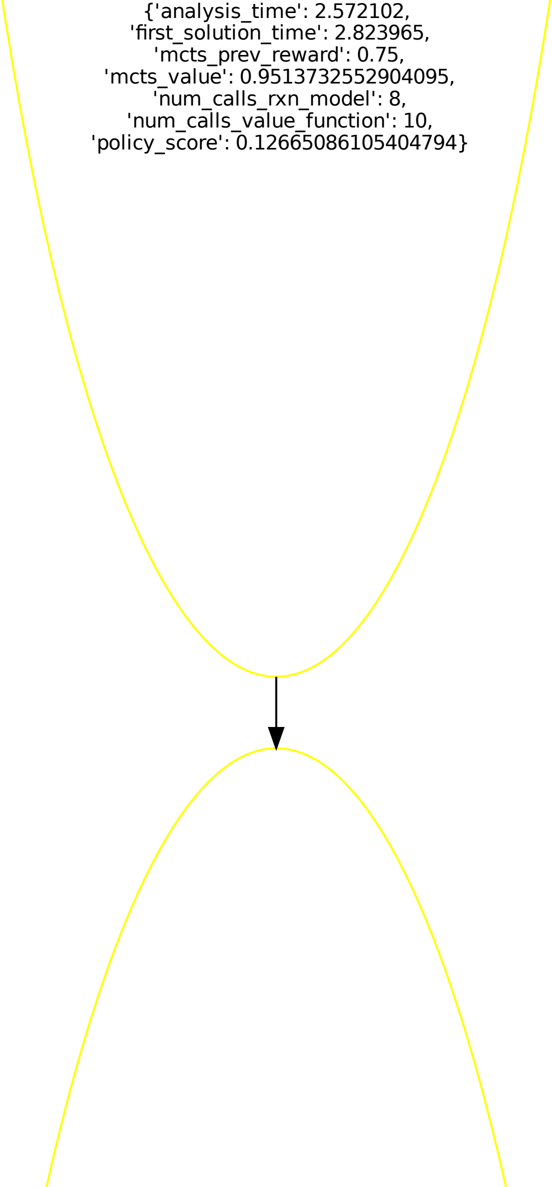


HCl

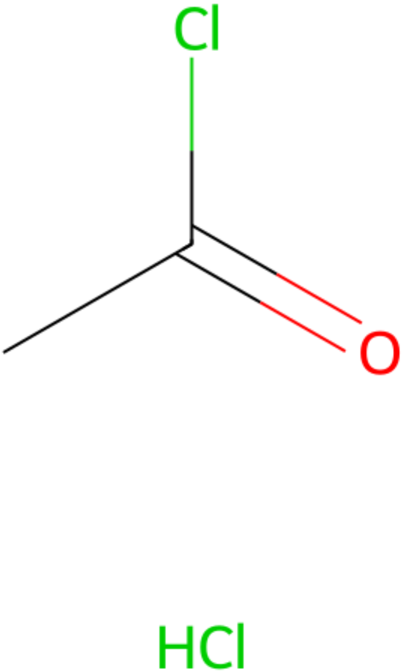






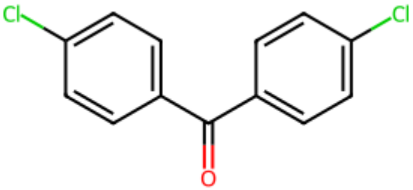


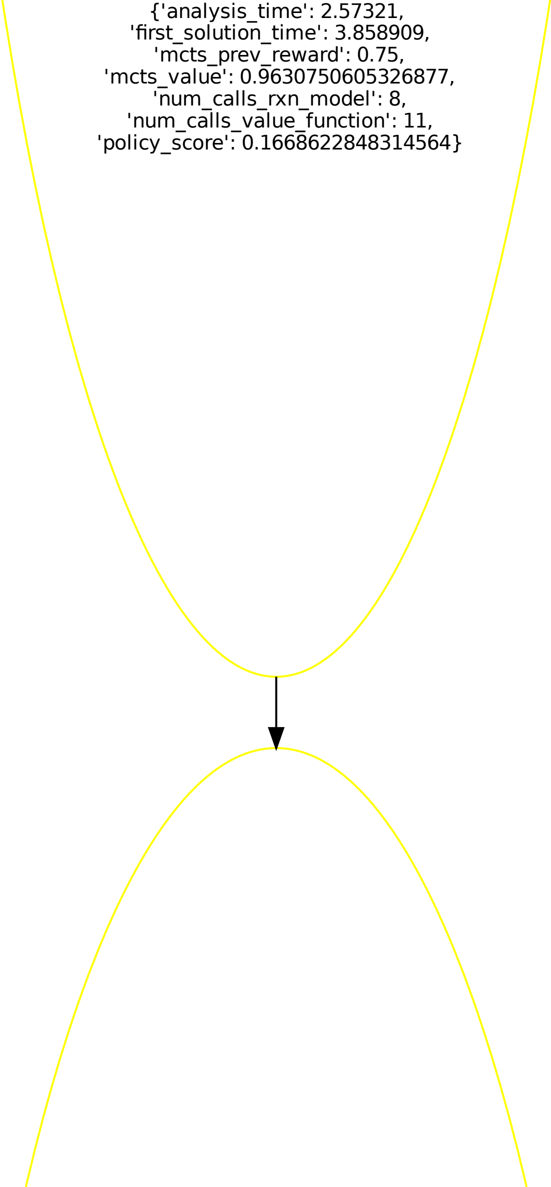
```
{'analysis_time': 2.572102,
'first_solution_time': 2.823965,
'mcts_prev_reward': 0.75,
'mcts_value': 0.9513732552904095,
'num_calls_rxn_model': 8,
'num_calls_value_function': 10,
'policy_score': 0.12665086105404794}
```



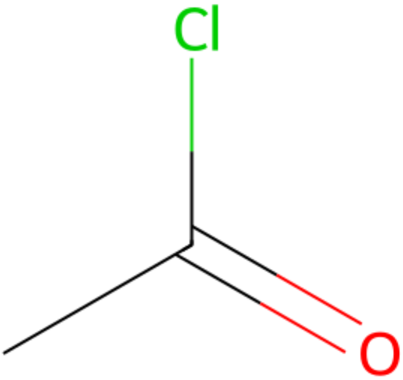
CH_4

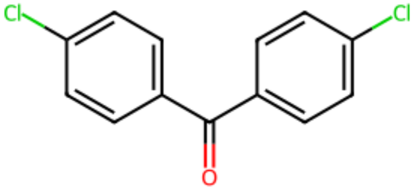
$Cl-Cl$





{'analysis_time': 2.57321,
'first_solution_time': 3.858909,
'mcts_prev_reward': 0.75,
'mcts_value': 0.9630750605326877,
'num_calls_rxn_model': 8,
'num_calls_value_function': 11,
'policy_score': 0.1668622848314564}

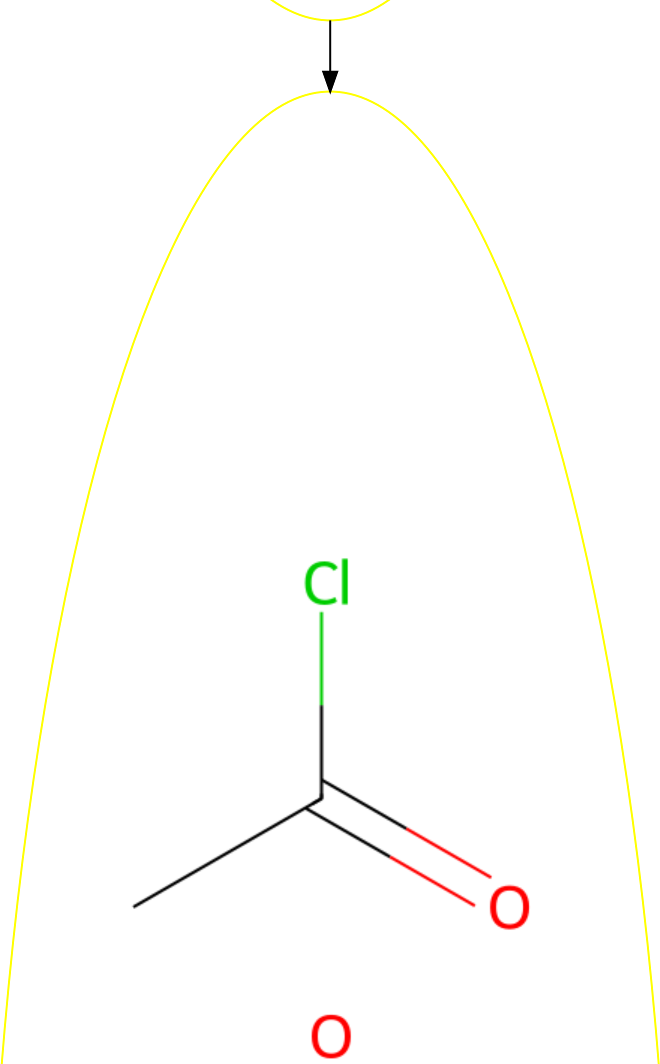


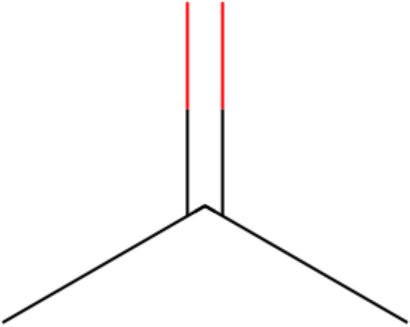


HCl

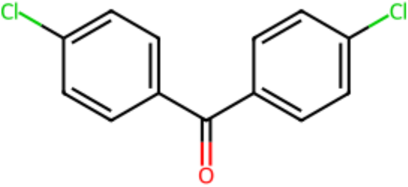
CH_4

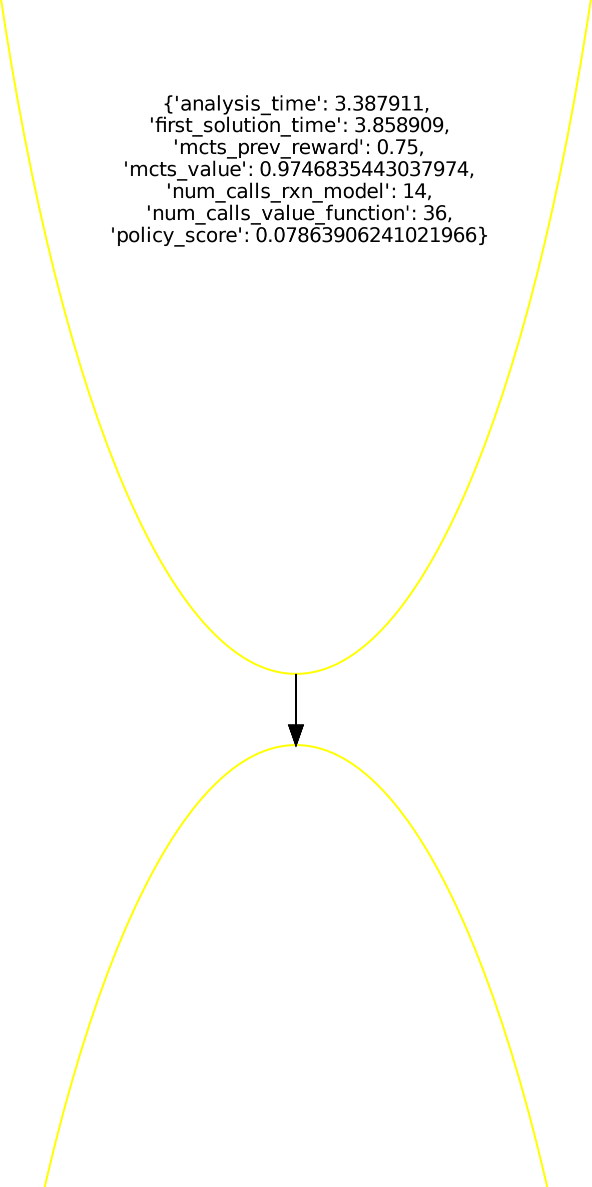
```
{'analysis_time': 3.384853,
'first_solution_time': 3.858909,
'mcts_prev_reward': 0.75,
'mcts_value': 0.9719687092568449,
'num_calls_rxn_model': 14,
'num_calls_value_function': 34,
'policy_score': 0.1490965091890023}
```



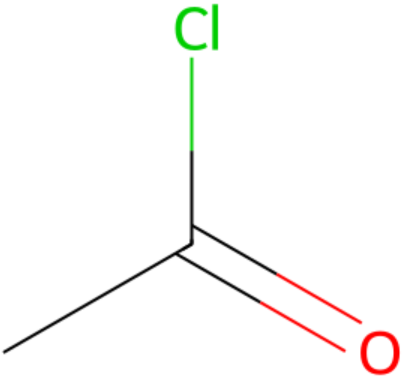


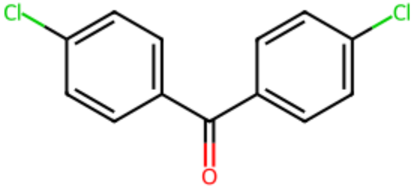
HCl



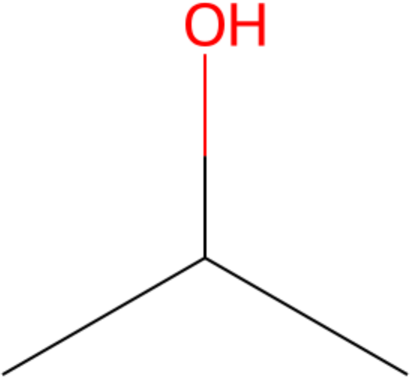


```
{'analysis_time': 3.387911,
'first_solution_time': 3.858909,
'mcts_prev_reward': 0.75,
'mcts_value': 0.9746835443037974,
'num_calls_rxn_model': 14,
'num_calls_value_function': 36,
'policy_score': 0.07863906241021966}
```

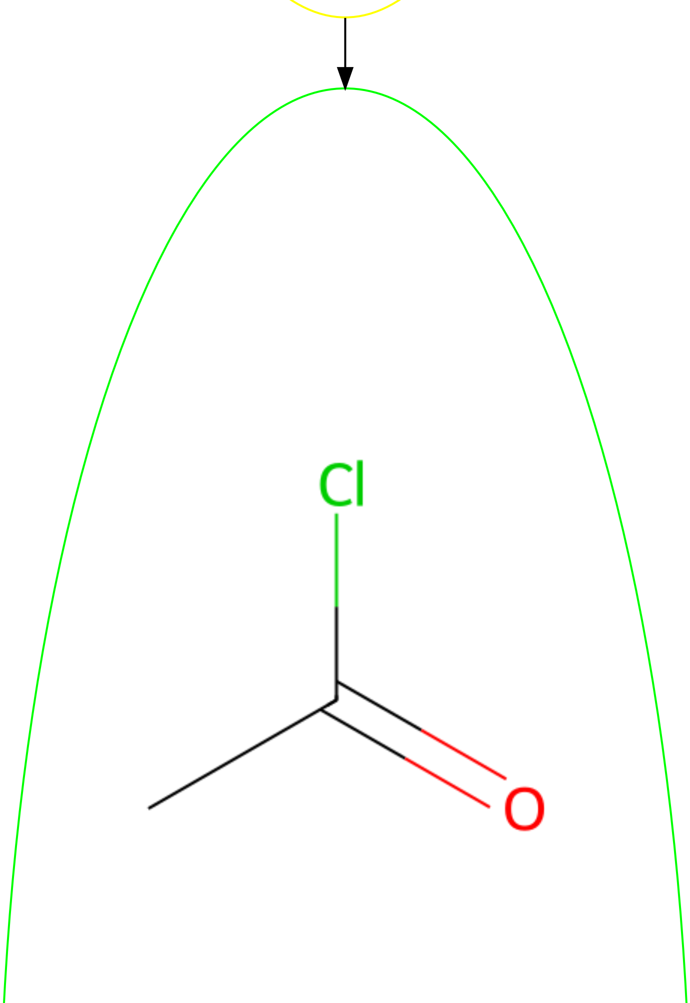


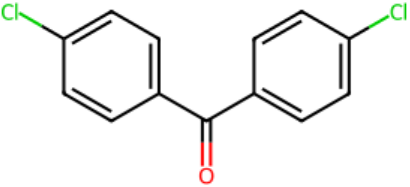


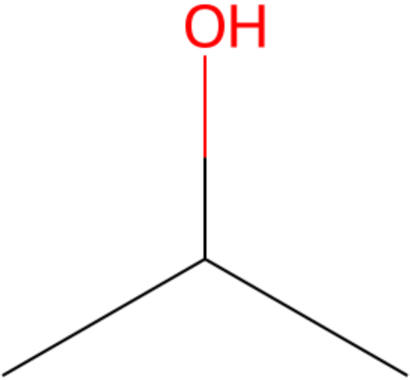
HCl



```
{'analysis_time': 3.393691,
'first_solution_time': 3.858909,
'mcts_prev_reward': 1.0,
'mcts_value': 0.9739583333333334,
'num_calls_rxn_model': 14,
'num_calls_value_function': 38,
'policy_score': 0.19416165661552667}
```







```
{'analysis_time': 3.858909,
'first_solution_time': 3.858909,
'mcts_prev_reward': 1.0,
'mcts_value': 1.0,
'num_calls_rxn_model': 18,
'num_calls_value_function': 62,
'policy_score': 0.18173873149644126}
```